

Standard Operating Procedure Vertical Band Saw

Band Saw

A band saw is an almost indispensable machine in a shop used for making curved cuts and re-sawing, or cutting wood/metal to make it thinner. The blade runs over wheels above and below a table, allowing wood to be cut where the blade is traveling downward through the table. The band saw is a rough cutting and shaping tool; it is not intended for finished cuts.

Personal Protective Equipment					
 **Tie Back Hair & Secure Loose Items**	 **Closed Toe Shoes**	 **Eye/Face Protection**	 **Hearing Protection**	 **Protective Clothing**	 **No Gloves Advised**
Tie back long hair and secure loose clothing that could get caught in rotating parts. Remove rings and other jewelry.	Appropriate enclosed footwear should be worn.	Wear ANSI Z87.1-compliant safety goggles to protect eyes. Face shields may be necessary to protect from dust, shards, and debris.	Hearing protection should be worn when noise levels are excessive.	Close-fitting or protective clothing, such as an apron, is encouraged.	Gloves can get caught in rotating parts. Use push sticks or boards and follow the "4 inch" rule to keep hands away from harm.

Potential Hazards

- Be aware of the following potential hazards when working with band saws:
  - **Mechanical:** points of operation, pinch points, shear points, power transmission points
  - **Operational:** heat, dust/chips/shards, noise, unstable loads/stocks
  - **Chemical:** coolants, cleaning solvents, lubricants
  - **Electrical/Energy Sources:** exposed wiring, malfunctioning equipment, unexpected start up/shut down

Before Use

- Know the location of start and stop switches or buttons, and keep the band saw area free of tools and other materials.



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- Report any unsafe conditions observed on the band saw or stock being worked on to the shop supervisor.
- Check the Safety Data Sheet for potential health hazards of specific materials (e.g., western red cedar, beryllium, chromium) and chemicals (e.g., lubricants, solvents).
- Ensure that machine guards are in place and recommended personal protective equipment is worn.
- Ensure the blade is properly adjusted prior to turning on the machine.
- Adjust the upper guard assembly to within ¼ inch of the stock prior to starting the machine.
- If adjustments or maintenance (such as adding oil) are needed, unplug the power (or lockout/tagout if hard-wired) before adjusting.
- Set the band saw at the appropriate speed for the type of stock being machined.
- Check to ensure the band saw blade is sharpened, and that the type of blade is correct for the type of stock and speed being used.
- NEVER shift to different gear at high RPM. Must drop RPM as low as possible.
- Be cautious when a cutting operation requires locating fingers close to the blade. It is a best practice to keep hands at least 4 inches away from the blade.

During Use

- Concentrate on the task at hand; avoid distractions. Most injuries occur when the operator is not paying full attention to what they are doing.
- Allow the saw to reach full set speed prior to cutting stock.
- Do not force stock into the saw blade. Let the speed of the blade cut stock appropriately.



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- Make “release” cuts before cutting long curves, and plan saw cuts to avoid backing out of curves in the stock.
- **Never** try to saw round stock (small logs or dowels) unless they are securely fastened in a V-block or special log jig (sled).
- Never push a piece of stock with hands in front of the saw blade. Use a push stick.
- Keep hands at a safe distance on either side of the stock being machined. Use a push stick or board to push small or irregularly sized pieces of stock. Small work pieces can also be secured with a tabletop vise or clamp.
- All round stock must be secured in a tabletop vise or clamp prior to starting the cut.
- Hold the stock flat on the table prior to starting the cut.
- If the saw blade binds on a piece of stock, turn the saw off and wait until it comes to a complete stop before attempting to remove the blade from the stock.
- Do not allow large quantities of chips to accumulate around the work piece or band saw table. Use a brush to remove excess chips.

After Use

- Shut off the power and/or unplug the machine and ensure that it is ready for the next person to use. Wait for the saw blade to completely stop on its own- do not attempt to use your hands to stop it from moving.
- Sweep or use a shop vacuum to collect any debris that resulted from cutting on the floor or other surfaces. Avoid cleaning with compressed air to prevent injury.
- Store stock materials in a neat and secured manner; do not accumulate excess combustibles. Keep aisles and exits clear.



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- Wash hands and exposed skin thoroughly after completing work and before leaving the shop area.

Emergency Equipment and Procedures

- Report all incidents to the shop supervisor, including injuries, equipment damage, and near misses.
- Know the location of the nearest emergency equipment and items, such as the emergency stop/shut-off for the specific machine, eye wash, first aid kit, fire extinguishers, fire alarm pull stations, and emergency exits.

